

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

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saireddy@saireddy:~/Practical$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M 1 disk
sdb   8:16   0 159.4M 1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

saireddy@saireddy:~/Practical$
saireddy@saireddy:~/Practical$ ps
  PID TTY          TIME CMD
 321 pts/0    00:00:00 bash
 575 pts/0    00:00:00 ps
saireddy@saireddy:~/Practical$
saireddy@saireddy:~/Practical$ top
top - 08:53:10 up 11 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 26 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.2 us,  0.1 sy,  0.0 ni, 99.6 id,  0.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7961.0 total, 5210.1 free, 1326.9 used, 1424.0 buff/cache.
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 6596.1 avail Mem.

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        0   0 23124 3324 888 S  0.5   0.2   0:01.69 systemd
    2 root        0   0    0    0  0 S  0.5   0.0   0:00.03 kthreadd
    4 root        0   0    0    0  0 S  0.1   0.0   0:00.07 kworker/0:60-events_highpri
    7 root        0   0    0    0  0 S  0.1   0.0   0:00.19 kworker/0:10-rcu_gp
   10 root        0   0    0    0  0 S  0.1   0.0   0:00.00 rcu_tasks_rude-
   11 root        0   0    0    0  0 S  0.1   0.0   0:00.00 rcu_tasks_trace
   12 root        0   0    0    0  0 S  0.1   0.0   0:00.31 ksoftirqd/0
   13 root        0   0    0    0  0 S  0.1   0.0   0:00.00 rcu_sched
   14 root       -20  0    0    0  0 S  0.1   0.0   0:00.00 migration/0
   15 root        0   0    0    0  0 S  0.1   0.0   0:00.20 idle_inject/0
   16 root        0   0    0    0  0 S  0.1   0.0   0:00.00 cpahp/0
   17 root        0   0    0    0  0 S  0.5   0.0   0:00.01 cpahp/1
   18 root        0   0    0    0  0 S  0.5   0.0   0:00.00 migration/1
   19 root        0   0    0    0  0 S  0.5   0.0   0:00.07 ksoftirqd/1
   20 root        0   0    0    0  0 S  0.1   0.0   0:00.10 kworker/1:0-events
   23 root        0   0    0    0  0 S  0.1   0.0   0:00.00 kdevtmpfs
   24 root        0 -20  0    0  0 S  0.1   0.0   0:00.00 netns
   25 root        0   0    0    0  0 S  0.1   0.0   0:00.09 kworker/1:1H-kblockd
   27 root        0   0    0    0  0 S  0.1   0.0   0:00.06 khungtaskd
   28 root        0   0    0    0  0 S  0.1   0.0   0:00.00 oom_reaper
 206 saireddy    20  0 23124 11880 7312 S  0.0   0.1   0:00.15 dbus-daemon
 315 saireddy    20  0 27192 16472 9572 S  0.0   0.2   0:00.16 systemd --user
 321 saireddy    20  0 31840 13564 9704 S  0.0   0.2   0:00.02 bash
 575 saireddy    20  0 3624 2980 2312 R  0.0   0.1   0:00.00 top
```

```

saireddy@saireddy:~$ mkdir OSLab
saireddy@saireddy:~$ mkdir Practical
saireddy@saireddy:~$ cd Practical
saireddy@saireddy:~/Practical$ nano Practical_1.c
saireddy@saireddy:~/Practical$ gcc Practical_1.c -o Practical_1
saireddy@saireddy:~/Practical$ ./Practical_1
Enter a command (e.g., ls, date, pwd): uname

[Parent Process] PID: 642 | Created Child PID: 643
[Parent Process] Waiting for child to finish...

[Child Process] PID: 643 | Parent PID: 642
[Child Process] Executing command: uname
Linux

[Parent Process] Child process completed. Exiting.
saireddy@saireddy:~/Practical$ |

saireddy@saireddy:~/Practical$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 5 5500U w/ Radeon 740M Graphics
Model:                 124
CPU family:             23
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               1
BogoMIPS:               6000.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep stttrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx
                        mmxext fxsr_opt pdpe1gb rdtscp rep_good nopl topology tsc_reliable nonstop_tsc cpuid extd_apicid
                        tsc_unchanged rtm pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 xsaveopt xsave cqm rdt_a rdseed
                        arat lbrv svm_lock 3bit bti pconfig smep smpl bts dtherm ida arat plbmgr tmm4b
                        hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topostat
                        perfctr_core ssbd sifrs lbrb stibp vmcall vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid avx512f
                        avx512dq rdseed adx smad avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt
                        xsavec xgetbv1 xsavec_etcstctrl avx512_bf16 clzero xsavecptr ace2 pku rpl rdpid save_tsc_scale
                        vms2_clean flushbyasid decodassist pausefilter optbreshold avx512_vnni umip avx512_vnni2 vnni gfn1
                        vaes vpmsuidq avx512_vnni avx512_bitalg avx512_vpopcntdq rdpid fsrcw

Virtualization features:
  Virtualization:      AMD-V
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                  192 MiB (6 instances)
  L1i:                  192 MiB (6 instances)
  L2:                   6 MiB (6 instances)
  L3:                   16 MiB (1 instance)
NUMA:
  NUMA node(s):         1
  NUMA node CPU(s):    0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect branch selection: Not affected
  ILL multihit:         Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Rbp_fls data sampling: Not affected
  Retbleed:             Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass:    Mitigation: Speculative Store Bypass disabled via prctl
  Spectre v1:           Mitigation: usercopy/swap barriers and __user pointer sanitization
  Spectre v2:           Mitigation: Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBHB-eIBRS Not affected; BHI Not affected
  Task async abort:     Not affected
  Weakswap:             Not affected
saireddy@saireddy:~/Practical$

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